Investigation workbook

Marketing A/B testing | exact SQL, complete outputs and interpretation

How to use this workbook

This is an audit trail, not just a list of queries. Each investigation records the question, exact SQLite SQL, every returned row, and an interpretation. Derived effect tables are produced by the accompanying Python code before SQL reads them. No SQL is invented after the fact.

Units and conventions

Rates in query outputs are proportions unless labeled lift_pp or ci_*_pp. One percentage point is 0.01 in probability units. PSA is control; product ads is treatment. Null signifies absent or undefined values. Exposure and peak-time fields are not assumed to be baseline covariates.

Source

https://www.kaggle.com/datasets/faviovaz/marketing-ab-testing

588,101 users; 14,843 conversions; zero excluded source rows. Downloaded 01 November 2026.

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01 / Source schema

Which fields were actually supplied?

```
PRAGMA table_info(raw_marketing);
```

Interpretation

Seven source columns include an exported index. The index is not a user characteristic or experimental variable.

cid	name	type	notnull	dflt value	pk
0	Unnamed: 0	INTEGER	0	NULL	0
1	user id	INTEGER	0	NULL	0
2	test group	TEXT	0	NULL	0
3	converted	INTEGER	0	NULL	0
4	total ads	INTEGER	0	NULL	0
5	most ads day	TEXT	0	NULL	0
6	most ads hour	INTEGER	0	NULL	0

02 / Source coverage

What is the analysis unit and how many outcomes exist?

```
SELECT COUNT(*) rows, COUNT(DISTINCT "user id") users,
SUM(converted) conversions FROM raw_marketing;
```

Interpretation

588,101 rows correspond to 588,101 distinct users and 14,843 conversions. No repeated-user aggregation is needed.

rows	users	conversions
588101	588101	14843

03 / Missingness

Are exclusions needed for missing source values?

```
SELECT SUM("user id" IS NULL) user_null,
SUM("test group" IS NULL) group_null, SUM(converted IS NULL) converted_null,
SUM("total ads" IS NULL) ads_null, SUM("most ads day" IS NULL) day_null,
SUM("most ads hour" IS NULL) hour_null FROM raw_marketing;
```

Interpretation

All six meaningful fields are complete. Zero rows are excluded for missingness.

user null	group null	converted null	ads null	day null	hour null
0	0	0	0	0	0

04 / Duplicate user IDs

Could repeated users violate the independent-user analysis?

```
SELECT COUNT(*) duplicate_users, COALESCE(SUM(n-1),0) extra_rows
FROM (SELECT user_id,COUNT(*) n FROM marketing GROUP BY user_id HAVING COUNT(*)>1);
```

Interpretation

No duplicate user identifiers were found. Independence still relies on user identity and assignment being correctly recorded; one row per ID cannot verify household or device interference.

duplicate users	extra rows
0	0

05 / Outcome and assignment categories

Are group labels and binary outcomes valid?

```
SELECT test_group,converted,COUNT(*) users FROM marketing
GROUP BY test_group,converted ORDER BY test_group,converted;
```

Interpretation

Only ad/psa labels and 0/1 outcomes occur. Keep both converters and non-converters in the denominator.

test group	converted	users
ad	0	550154
ad	1	14423
psa	0	23104
psa	1	420

06 / Range checks

Do exposure and time values fit their definitions?

```
SELECT MIN(total_ads) min_ads,MAX(total_ads) max_ads,
MIN(peak_hour) min_hour,MAX(peak_hour) max_hour,
SUM(total_ads<0 OR peak_hour NOT BETWEEN 0 AND 23) invalid_rows
FROM marketing;
```

Interpretation

Ads range from 1 to 2,065 and hours from 0 to 23. Large exposure counts are not automatically errors; they remain in the primary comparison.

min ads	max ads	min hour	max hour	invalid rows
1	2065	0	23	0

07 / Index audit

Is the extra index a usable analytical variable?

```
SELECT COUNT(*) rows,COUNT(DISTINCT "Unnamed: 0") distinct_index,
MIN("Unnamed: 0") first_index,MAX("Unnamed: 0") last_index
FROM raw_marketing;
```

Interpretation

The index runs from 0 through 588,100. It is dropped during cleaning; there is no evidence it represents time or ordering of assignment.

rows	distinct index	first index	last index
588101	588101	0	588100

08 / Cleaning reconciliation

Did preparation change outcomes or cohort size?

```
SELECT COUNT(*) users,COUNT(DISTINCT user_id) distinct_users,
SUM(converted) conversions,SUM(total_ads) impressions FROM marketing;
```

Interpretation

All original users and outcomes are retained. Preparation renames fields, converts the Boolean outcome to 0/1, removes the index and adds derived labels. No outliers are discarded.

users	distinct users	conversions	impressions
588101	588101	14843	14597182

09 / Allocation

How unequal are the observed arm sizes?

```
SELECT test_group,COUNT(*) users,
1.0*COUNT(*)/(SELECT COUNT(*) FROM marketing) user_share
FROM marketing GROUP BY test_group;
```

Interpretation

Ad users represent 96.0000% of observations. Unequal allocation is not itself evidence of failure. A target split must be specified independently; 96/4 is only a diagnostic scenario here.

test group	users	user share
ad	564577	0.96
psa	23524	0.0399999

10 / Primary outcomes

What are the conversion rates and correct denominators?

```
SELECT test_group,COUNT(*) users,SUM(converted) conversions,
AVG(converted) conversion_rate FROM marketing GROUP BY test_group;
```

Interpretation

The ad-minus-PSA difference is 0.7692 percentage points, or 43.09% relative lift. The smaller PSA arm governs most of the precision.

test group	users	conversions	conversion rate
ad	564577	14423	0.0255466
psa	23524	420	0.0178541

11 / Exposure volume

How much exposure is associated with each arm?

```
SELECT test_group,SUM(total_ads) impressions,AVG(total_ads) ads_per_user,
AVG(CASE WHEN converted=1 THEN total_ads END) ads_per_converter,
AVG(CASE WHEN converted=0 THEN total_ads END) ads_per_nonconverter
FROM marketing GROUP BY test_group;
```

Interpretation

These are descriptive exposure summaries. Exposure may be affected by treatment, behavior and conversion; it cannot safely serve as a pre-treatment adjustment variable.

test group	impressions	ads per user	ads per converter	ads per nonconverter
ad	14014701	24.8234	83.9056	23.2744
psa	582481	24.7611	83.2762	23.6974

12 / Frequency-band outcomes

How does conversion vary with observed exposure?

```
SELECT frequency_band,test_group,COUNT(*) users,SUM(converted) conversions,
AVG(converted) rate,SUM(total_ads) impressions FROM marketing
GROUP BY frequency_band,test_group ORDER BY frequency_band,test_group;
```

Interpretation

These bands were selected for readability, not optimized for significance. Different exposure bands contain different kinds of users; apparent dose-response does not establish a causal frequency effect.

frequency band	test group	users	conversions	rate	impressions
01 | 1-5	ad	169962	427	0.00251233	441828
01 | 1-5	psa	7861	22	0.00279863	20634
02 | 6-10	ad	79537	386	0.00485309	608167
02 | 6-10	psa	3415	23	0.00673499	25812
03 | 11-20	ad	123334	1036	0.00839995	1885148
03 | 11-20	psa	4150	34	0.00819277	63590
04 | 21-50	ad	125541	3660	0.0291538	3959033
04 | 21-50	psa	5235	114	0.0217765	162262
05 | 51-100	ad	44149	5135	0.116311	3062470
05 | 51-100	psa	1853	107	0.0577442	129374
06 | 101-200	ad	16360	2892	0.176773	2210760
06 | 101-200	psa	752	80	0.106383	102652
07 | 201+	ad	5694	887	0.155778	1847295
07 | 201+	psa	258	40	0.155039	78157

13 / Weekday outcomes

How do rates vary by peak-exposure weekday?

```
SELECT day_order,peak_day,test_group,COUNT(*) users,
SUM(converted) conversions,AVG(converted) rate FROM marketing
GROUP BY day_order,peak_day,test_group ORDER BY day_order,test_group;
```

Interpretation

Weekday means the day with the most ad exposure for that user. It is not the day of conversion, a dated cohort or a randomized scheduling treatment.

day order	peak day	test group	users	conversions	rate
1	Monday	ad	83571	2778	0.0332412
1	Monday	psa	3502	79	0.0225585
2	Tuesday	ad	74572	2270	0.0304404
2	Tuesday	psa	2907	42	0.0144479
3	Wednesday	ad	77418	1963	0.0253559
3	Wednesday	psa	3490	55	0.0157593
4	Thursday	ad	79077	1711	0.0216371
4	Thursday	psa	3905	79	0.0202305
5	Friday	ad	88805	1995	0.022465
5	Friday	psa	3803	62	0.0163029
6	Saturday	ad	78802	1679	0.0213066
6	Saturday	psa	2858	40	0.0139958
7	Sunday	ad	82332	2027	0.0246198
7	Sunday	psa	3059	63	0.020595

14 / Hour outcomes

What is the observed peak-hour conversion profile?

```
SELECT peak_hour,test_group,COUNT(*) users,SUM(converted) conversions,
AVG(converted) rate FROM marketing GROUP BY peak_hour,test_group
ORDER BY peak_hour,test_group;
```

Interpretation

Use counts with rates. Sparse overnight PSA cells have noisy estimates, and selecting the highest observed hour would overstate certainty.

peak hour	test group	users	conversions	rate
0	ad	5309	102	0.0192127
0	psa	227	0	0
1	ad	4615	62	0.0134345
1	psa	187	0	0
2	ad	5152	39	0.00756988
2	psa	181	0	0
3	ad	2590	27	0.0104247
3	psa	89	1	0.011236
4	ad	694	11	0.0158501
4	psa	28	0	0
5	ad	742	16	0.0215633
5	psa	23	0	0
6	ad	1985	46	0.0231738
6	psa	83	0	0
7	ad	6168	114	0.0184825
7	psa	237	2	0.00843882
8	ad	16968	337	0.0198609
8	psa	659	7	0.0106222
9	ad	29802	582	0.0195289
9	psa	1202	13	0.0108153
10	ad	37454	818	0.0218401
10	psa	1485	20	0.013468

peak hour	test group	users	conversions	rate
11	ad	44149	992	0.0224694
11	psa	2061	30	0.014556
12	ad	45238	1092	0.024139
12	psa	2060	35	0.0169903
13	ad	45485	1140	0.0250632
13	psa	2170	36	0.0165899
14	ad	43779	1251	0.0285753
14	psa	1869	30	0.0160514
15	ad	42855	1279	0.0298448
15	psa	1828	46	0.0251641
16	ad	35963	1111	0.0308929
16	psa	1604	45	0.0280549
17	ad	33605	959	0.0285374
17	psa	1383	28	0.0202458
18	ad	31052	853	0.0274701
18	psa	1271	32	0.025177
19	ad	29169	782	0.0268093
19	psa	1183	29	0.0245139
20	ad	27846	843	0.0302736
20	psa	1077	19	0.0176416
21	ad	28895	843	0.0291746
21	psa	1081	24	0.0222017
22	ad	25515	675	0.026455
22	psa	917	15	0.0163577
23	ad	19547	449	0.0229703
23	psa	619	8	0.0129241

15 / Day-hour cells

What data support the heatmap?

```
SELECT day_order,peak_day,peak_hour,test_group,COUNT(*) users,
SUM(converted) conversions,AVG(converted) rate FROM marketing
GROUP BY day_order,peak_day,peak_hour,test_group
ORDER BY day_order,peak_hour,test_group;
```

Interpretation

All populated day-hour-arm cells are shown. The report heatmap uses ad-arm conversion rates; it is an exploratory exposure-context display, not a scheduling recommendation.

day order	peak day	peak hour	test group	users	conversions	rate
1	Monday	0	ad	820	14	0.0170732
1	Monday	0	psa	42	0	0
1	Monday	1	ad	531	9	0.0169492
1	Monday	1	psa	35	0	0
1	Monday	2	ad	709	5	0.00705219
1	Monday	2	psa	28	0	0
1	Monday	3	ad	316	4	0.0126582
1	Monday	3	psa	14	0	0
1	Monday	4	ad	88	0	0
1	Monday	4	psa	4	0	0
1	Monday	5	ad	92	3	0.0326087
1	Monday	5	psa	2	0	0
1	Monday	6	ad	262	11	0.0419847
1	Monday	6	psa	17	0	0
1	Monday	7	ad	760	14	0.0184211
1	Monday	7	psa	26	0	0
1	Monday	8	ad	2412	82	0.0339967
1	Monday	8	psa	98	0	0
1	Monday	9	ad	4365	116	0.026575
1	Monday	9	psa	181	3	0.0165746
1	Monday	10	ad	5159	157	0.0304323
1	Monday	10	psa	229	6	0.0262009

day order	peak day	peak hour	test group	users	conversions	rate
1	Monday	11	ad	6433	210	0.0326442
1	Monday	11	psa	287	8	0.0278746
1	Monday	12	ad	6744	224	0.0332147
1	Monday	12	psa	258	7	0.0271318
1	Monday	13	ad	6678	232	0.0347409
1	Monday	13	psa	292	6	0.0205479
1	Monday	14	ad	6039	271	0.044875
1	Monday	14	psa	296	10	0.0337838
1	Monday	15	ad	6357	273	0.0429448
1	Monday	15	psa	327	11	0.0336391
1	Monday	16	ad	5592	209	0.0373748
1	Monday	16	psa	266	8	0.0300752
1	Monday	17	ad	5432	195	0.0358984
1	Monday	17	psa	193	6	0.0310881
1	Monday	18	ad	5038	140	0.0277888
1	Monday	18	psa	182	3	0.0164835
1	Monday	19	ad	4449	136	0.0305687
1	Monday	19	psa	171	6	0.0350877
1	Monday	20	ad	4567	148	0.0324064
1	Monday	20	psa	158	2	0.0126582
1	Monday	21	ad	4244	144	0.0339303
1	Monday	21	psa	159	1	0.00628931
1	Monday	22	ad	3934	118	0.0299949
1	Monday	22	psa	149	2	0.0134228
1	Monday	23	ad	2550	63	0.0247059
1	Monday	23	psa	88	0	0
2	Tuesday	0	ad	713	14	0.0196353
2	Tuesday	0	psa	24	0	0
2	Tuesday	1	ad	632	9	0.0142405
2	Tuesday	1	psa	20	0	0

day order	peak day	peak hour	test group	users	conversions	rate
2	Tuesday	2	ad	728	4	0.00549451
2	Tuesday	2	psa	16	0	0
2	Tuesday	3	ad	336	7	0.0208333
2	Tuesday	3	psa	7	1	0.142857
2	Tuesday	4	ad	100	2	0.02
2	Tuesday	4	psa	4	0	0
2	Tuesday	5	ad	76	1	0.0131579
2	Tuesday	6	ad	221	5	0.0226244
2	Tuesday	6	psa	4	0	0
2	Tuesday	7	ad	729	16	0.0219479
2	Tuesday	7	psa	22	0	0
2	Tuesday	8	ad	2412	44	0.0182421
2	Tuesday	8	psa	99	0	0
2	Tuesday	9	ad	4440	105	0.0236486
2	Tuesday	9	psa	183	2	0.010929
2	Tuesday	10	ad	5773	122	0.0211329
2	Tuesday	10	psa	233	4	0.0171674
2	Tuesday	11	ad	6481	176	0.0271563
2	Tuesday	11	psa	361	2	0.00554017
2	Tuesday	12	ad	6588	205	0.0311172
2	Tuesday	12	psa	273	3	0.010989
2	Tuesday	13	ad	6491	210	0.0323525
2	Tuesday	13	psa	306	3	0.00980392
2	Tuesday	14	ad	6371	230	0.0361011
2	Tuesday	14	psa	247	3	0.0121457
2	Tuesday	15	ad	5656	233	0.0411952
2	Tuesday	15	psa	216	4	0.0185185
2	Tuesday	16	ad	4103	187	0.0455764
2	Tuesday	16	psa	159	6	0.0377358
2	Tuesday	17	ad	4084	144	0.0352595

day order	peak day	peak hour	test group	users	conversions	rate
2	Tuesday	17	psa	160	1	0.00625
2	Tuesday	18	ad	3621	131	0.0361779
2	Tuesday	18	psa	120	4	0.0333333
2	Tuesday	19	ad	3298	94	0.0285021
2	Tuesday	19	psa	90	1	0.0111111
2	Tuesday	20	ad	3397	92	0.0270827
2	Tuesday	20	psa	90	2	0.0222222
2	Tuesday	21	ad	3898	93	0.0238584
2	Tuesday	21	psa	145	4	0.0275862
2	Tuesday	22	ad	2639	91	0.0344828
2	Tuesday	22	psa	99	1	0.010101
2	Tuesday	23	ad	1785	55	0.0308123
2	Tuesday	23	psa	29	1	0.0344828
3	Wednesday	0	ad	588	7	0.0119048
3	Wednesday	0	psa	10	0	0
3	Wednesday	1	ad	596	5	0.00838926
3	Wednesday	1	psa	12	0	0
3	Wednesday	2	ad	619	3	0.00484653
3	Wednesday	2	psa	20	0	0
3	Wednesday	3	ad	330	1	0.0030303
3	Wednesday	3	psa	8	0	0
3	Wednesday	4	ad	100	2	0.02
3	Wednesday	4	psa	2	0	0
3	Wednesday	5	ad	140	1	0.00714286
3	Wednesday	5	psa	3	0	0
3	Wednesday	6	ad	358	5	0.0139665
3	Wednesday	6	psa	7	0	0
3	Wednesday	7	ad	840	10	0.0119048
3	Wednesday	7	psa	17	0	0
3	Wednesday	8	ad	2287	38	0.0166157

day order	peak day	peak hour	test group	users	conversions	rate
3	Wednesday	8	psa	64	0	0
3	Wednesday	9	ad	4625	79	0.0170811
3	Wednesday	9	psa	154	1	0.00649351
3	Wednesday	10	ad	5812	148	0.0254646
3	Wednesday	10	psa	166	1	0.0060241
3	Wednesday	11	ad	6852	167	0.0243724
3	Wednesday	11	psa	274	3	0.0109489
3	Wednesday	12	ad	6938	178	0.0256558
3	Wednesday	12	psa	367	6	0.0163488
3	Wednesday	13	ad	6508	165	0.0253534
3	Wednesday	13	psa	368	4	0.0108696
3	Wednesday	14	ad	5372	160	0.0297841
3	Wednesday	14	psa	231	4	0.017316
3	Wednesday	15	ad	5734	172	0.0299965
3	Wednesday	15	psa	277	5	0.0180505
3	Wednesday	16	ad	4616	165	0.0357452
3	Wednesday	16	psa	216	8	0.037037
3	Wednesday	17	ad	4334	140	0.0323027
3	Wednesday	17	psa	228	3	0.0131579
3	Wednesday	18	ad	4073	112	0.0274982
3	Wednesday	18	psa	246	7	0.0284553
3	Wednesday	19	ad	3894	81	0.0208012
3	Wednesday	19	psa	215	4	0.0186047
3	Wednesday	20	ad	3290	85	0.0258359
3	Wednesday	20	psa	204	4	0.0196078
3	Wednesday	21	ad	3351	92	0.0274545
3	Wednesday	21	psa	153	2	0.0130719
3	Wednesday	22	ad	3277	83	0.025328
3	Wednesday	22	psa	157	3	0.0191083
3	Wednesday	23	ad	2884	64	0.0221914

day order	peak day	peak hour	test group	users	conversions	rate
3	Wednesday	23	psa	91	0	0
4	Thursday	0	ad	708	18	0.0254237
4	Thursday	0	psa	44	0	0
4	Thursday	1	ad	691	13	0.0188133
4	Thursday	1	psa	34	0	0
4	Thursday	2	ad	683	9	0.0131772
4	Thursday	2	psa	39	0	0
4	Thursday	3	ad	363	2	0.00550964
4	Thursday	3	psa	8	0	0
4	Thursday	4	ad	109	3	0.0275229
4	Thursday	4	psa	6	0	0
4	Thursday	5	ad	91	2	0.021978
4	Thursday	5	psa	5	0	0
4	Thursday	6	ad	285	6	0.0210526
4	Thursday	6	psa	18	0	0
4	Thursday	7	ad	1005	16	0.0159204
4	Thursday	7	psa	60	0	0
4	Thursday	8	ad	2722	41	0.0150625
4	Thursday	8	psa	140	4	0.0285714
4	Thursday	9	ad	4404	70	0.0158946
4	Thursday	9	psa	220	4	0.0181818
4	Thursday	10	ad	4945	85	0.0171891
4	Thursday	10	psa	233	1	0.00429185
4	Thursday	11	ad	5171	118	0.0228196
4	Thursday	11	psa	283	5	0.0176678
4	Thursday	12	ad	5892	115	0.019518
4	Thursday	12	psa	317	7	0.022082
4	Thursday	13	ad	6374	124	0.019454
4	Thursday	13	psa	368	5	0.013587
4	Thursday	14	ad	6543	167	0.0255235

day order	peak day	peak hour	test group	users	conversions	rate
4	Thursday	14	psa	323	8	0.0247678
4	Thursday	15	ad	5980	147	0.0245819
4	Thursday	15	psa	282	8	0.0283688
4	Thursday	16	ad	5523	145	0.0262538
4	Thursday	16	psa	263	9	0.0342205
4	Thursday	17	ad	4801	103	0.0214539
4	Thursday	17	psa	232	4	0.0172414
4	Thursday	18	ad	4303	98	0.0227748
4	Thursday	18	psa	205	5	0.0243902
4	Thursday	19	ad	3970	96	0.0241814
4	Thursday	19	psa	193	3	0.015544
4	Thursday	20	ad	3801	109	0.0286767
4	Thursday	20	psa	178	3	0.0168539
4	Thursday	21	ad	4222	88	0.0208432
4	Thursday	21	psa	175	4	0.0228571
4	Thursday	22	ad	3680	80	0.0217391
4	Thursday	22	psa	160	6	0.0375
4	Thursday	23	ad	2811	56	0.0199217
4	Thursday	23	psa	119	3	0.0252101
5	Friday	0	ad	807	16	0.0198265
5	Friday	0	psa	34	0	0
5	Friday	1	ad	662	7	0.010574
5	Friday	1	psa	29	0	0
5	Friday	2	ad	825	10	0.0121212
5	Friday	2	psa	25	0	0
5	Friday	3	ad	515	2	0.0038835
5	Friday	3	psa	27	0	0
5	Friday	4	ad	118	2	0.0169492
5	Friday	4	psa	4	0	0
5	Friday	5	ad	129	2	0.0155039

day order	peak day	peak hour	test group	users	conversions	rate
5	Friday	5	psa	6	0	0
5	Friday	6	ad	330	4	0.0121212
5	Friday	6	psa	13	0	0
5	Friday	7	ad	1018	19	0.018664
5	Friday	7	psa	54	1	0.0185185
5	Friday	8	ad	2358	39	0.0165394
5	Friday	8	psa	110	1	0.00909091
5	Friday	9	ad	4282	86	0.0200841
5	Friday	9	psa	200	2	0.01
5	Friday	10	ad	5353	124	0.0231646
5	Friday	10	psa	223	5	0.0224215
5	Friday	11	ad	6927	124	0.017901
5	Friday	11	psa	356	5	0.0140449
5	Friday	12	ad	6949	136	0.0195712
5	Friday	12	psa	370	5	0.0135135
5	Friday	13	ad	7166	166	0.0231649
5	Friday	13	psa	328	3	0.00914634
5	Friday	14	ad	6622	163	0.0246149
5	Friday	14	psa	310	1	0.00322581
5	Friday	15	ad	6650	176	0.0264662
5	Friday	15	psa	285	6	0.0210526
5	Friday	16	ad	5523	153	0.0277023
5	Friday	16	psa	278	3	0.0107914
5	Friday	17	ad	5027	108	0.021484
5	Friday	17	psa	205	5	0.0243902
5	Friday	18	ad	4839	116	0.0239719
5	Friday	18	psa	203	6	0.0295567
5	Friday	19	ad	4593	95	0.0206836
5	Friday	19	psa	180	7	0.0388889
5	Friday	20	ad	4465	135	0.0302352

day order	peak day	peak hour	test group	users	conversions	rate
5	Friday	20	psa	133	2	0.0150376
5	Friday	21	ad	4933	144	0.0291912
5	Friday	21	psa	196	7	0.0357143
5	Friday	22	ad	4737	100	0.0211104
5	Friday	22	psa	135	0	0
5	Friday	23	ad	3977	68	0.0170983
5	Friday	23	psa	99	3	0.030303
6	Saturday	0	ad	763	18	0.0235911
6	Saturday	0	psa	28	0	0
6	Saturday	1	ad	819	7	0.00854701
6	Saturday	1	psa	27	0	0
6	Saturday	2	ad	900	4	0.00444444
6	Saturday	2	psa	31	0	0
6	Saturday	3	ad	362	4	0.0110497
6	Saturday	3	psa	12	0	0
6	Saturday	4	ad	79	0	0
6	Saturday	4	psa	6	0	0
6	Saturday	5	ad	88	5	0.0568182
6	Saturday	5	psa	6	0	0
6	Saturday	6	ad	252	13	0.0515873
6	Saturday	6	psa	13	0	0
6	Saturday	7	ad	727	17	0.0233838
6	Saturday	7	psa	27	1	0.037037
6	Saturday	8	ad	2150	50	0.0232558
6	Saturday	8	psa	81	1	0.0123457
6	Saturday	9	ad	3645	60	0.0164609
6	Saturday	9	psa	129	0	0
6	Saturday	10	ad	4867	79	0.0162318
6	Saturday	10	psa	179	0	0
6	Saturday	11	ad	5760	84	0.0145833

day order	peak day	peak hour	test group	users	conversions	rate
6	Saturday	11	psa	247	5	0.0202429
6	Saturday	12	ad	5852	108	0.0184552
6	Saturday	12	psa	252	4	0.015873
6	Saturday	13	ad	6056	121	0.0199802
6	Saturday	13	psa	238	8	0.0336134
6	Saturday	14	ad	5849	118	0.0201744
6	Saturday	14	psa	209	1	0.00478469
6	Saturday	15	ad	5869	128	0.0218095
6	Saturday	15	psa	222	4	0.018018
6	Saturday	16	ad	5223	104	0.0199119
6	Saturday	16	psa	223	4	0.0179372
6	Saturday	17	ad	4850	109	0.0224742
6	Saturday	17	psa	166	3	0.0180723
6	Saturday	18	ad	4368	112	0.025641
6	Saturday	18	psa	143	2	0.013986
6	Saturday	19	ad	4445	119	0.0267717
6	Saturday	19	psa	160	2	0.0125
6	Saturday	20	ad	4593	111	0.0241672
6	Saturday	20	psa	155	1	0.00645161
6	Saturday	21	ad	4542	146	0.0321444
6	Saturday	21	psa	117	3	0.025641
6	Saturday	22	ad	3904	99	0.0253586
6	Saturday	22	psa	93	0	0
6	Saturday	23	ad	2839	63	0.0221909
6	Saturday	23	psa	94	1	0.0106383
7	Sunday	0	ad	910	15	0.0164835
7	Sunday	0	psa	45	0	0
7	Sunday	1	ad	684	12	0.0175439
7	Sunday	1	psa	30	0	0
7	Sunday	2	ad	688	4	0.00581395

day order	peak day	peak hour	test group	users	conversions	rate
7	Sunday	2	psa	22	0	0
7	Sunday	3	ad	368	7	0.0190217
7	Sunday	3	psa	13	0	0
7	Sunday	4	ad	100	2	0.02
7	Sunday	4	psa	2	0	0
7	Sunday	5	ad	126	2	0.015873
7	Sunday	5	psa	1	0	0
7	Sunday	6	ad	277	2	0.00722022
7	Sunday	6	psa	11	0	0
7	Sunday	7	ad	1089	22	0.020202
7	Sunday	7	psa	31	0	0
7	Sunday	8	ad	2627	43	0.0163685
7	Sunday	8	psa	67	1	0.0149254
7	Sunday	9	ad	4041	66	0.0163326
7	Sunday	9	psa	135	1	0.00740741
7	Sunday	10	ad	5545	103	0.0185753
7	Sunday	10	psa	222	3	0.0135135
7	Sunday	11	ad	6525	113	0.017318
7	Sunday	11	psa	253	2	0.00790514
7	Sunday	12	ad	6275	126	0.0200797
7	Sunday	12	psa	223	3	0.0134529
7	Sunday	13	ad	6212	122	0.0196394
7	Sunday	13	psa	270	7	0.0259259
7	Sunday	14	ad	6983	142	0.0203351
7	Sunday	14	psa	253	3	0.0118577
7	Sunday	15	ad	6609	150	0.0226963
7	Sunday	15	psa	219	8	0.0365297
7	Sunday	16	ad	5383	148	0.027494
7	Sunday	16	psa	199	7	0.0351759
7	Sunday	17	ad	5077	160	0.0315147

day order	peak day	peak hour	test group	users	conversions	rate
7	Sunday	17	psa	199	6	0.0301508
7	Sunday	18	ad	4810	144	0.0299376
7	Sunday	18	psa	172	5	0.0290698
7	Sunday	19	ad	4520	161	0.0356195
7	Sunday	19	psa	174	6	0.0344828
7	Sunday	20	ad	3733	163	0.0436646
7	Sunday	20	psa	159	5	0.0314465
7	Sunday	21	ad	3705	136	0.0367072
7	Sunday	21	psa	136	3	0.0220588
7	Sunday	22	ad	3344	104	0.0311005
7	Sunday	22	psa	124	3	0.0241935
7	Sunday	23	ad	2701	80	0.0296187
7	Sunday	23	psa	99	0	0

16 / Sparse day-hour cells

Where should heatmap comparisons be treated cautiously?

```
SELECT test_group,COUNT(*) cells,SUM(n<100) cells_under_100_users,
SUM(c<10) cells_under_10_conversions FROM
(SELECT test_group,peak_day,peak_hour,COUNT(*) n,SUM(converted) c
FROM marketing GROUP BY test_group,peak_day,peak_hour) GROUP BY test_group;
```

Interpretation

Low conversion counts, especially in the PSA arm, make fine-grained contrasts unstable. No 168-cell significance search is used to recommend a winning slot.

test group	cells	cells under 100 users	cells under 10 conversions
ad	168	6	38
psa	167	70	165

17 / Extreme exposures

How concentrated is the long tail?

```
SELECT test_group,COUNT(*) users,SUM(total_ads) impressions,
SUM(converted) conversions FROM marketing WHERE total_ads>202.0
GROUP BY test_group;
```

Interpretation

Above the common 99th-percentile threshold (202 exposures), 5,810 users account for 12.99% of all recorded impressions. Retain them for the primary test.

test group	users	impressions	conversions
ad	5560	1820290	869
psa	250	76545	40

18 / Exposure-trim sensitivity

Does the sign persist after a common tail exclusion?

```
SELECT test_group,COUNT(*) users,SUM(converted) conversions,
AVG(converted) rate FROM marketing WHERE total_ads<=202.0
GROUP BY test_group;
```

Interpretation

The trimmed difference is 0.7919 pp. This changes the population using a post-assignment variable; it is an influence diagnostic and cannot replace the full-cohort estimate.

test group	users	conversions	rate
ad	559017	13554	0.0242461
psa	23274	380	0.0163272

19 / Outcome by exposure band share

How does the exposure mix differ between arms?

```
SELECT frequency_band,test_group,COUNT(*) users,
1.0*COUNT(*)/SUM(COUNT(*)) OVER(PARTITION BY test_group) within_arm_share
FROM marketing GROUP BY frequency_band,test_group ORDER BY frequency_band,test_group;
```

Interpretation

Mix differences can explain descriptive patterns but are not baseline randomization balance checks. No pre-treatment covariates are supplied.

frequency band	test group	users	within arm share
01 | 1-5	ad	169962	0.301043
01 | 1-5	psa	7861	0.334169
02 | 6-10	ad	79537	0.140879
02 | 6-10	psa	3415	0.145171
03 | 11-20	ad	123334	0.218454
03 | 11-20	psa	4150	0.176416
04 | 21-50	ad	125541	0.222363
04 | 21-50	psa	5235	0.222539
05 | 51-100	ad	44149	0.0781984
05 | 51-100	psa	1853	0.0787706
06 | 101-200	ad	16360	0.0289774
06 | 101-200	psa	752	0.0319674
07 | 201+	ad	5694	0.0100854
07 | 201+	psa	258	0.0109675

20 / Counterfactual conversion calculation

What would the ad arm produce at the PSA rate?

```
WITH a AS (SELECT COUNT(*) n,SUM(converted) c FROM marketing WHERE test_group='ad'),
p AS (SELECT AVG(converted) rate FROM marketing WHERE test_group='psa')
SELECT a.n ad_users,a.c observed_ad_conversions,p.rate psa_rate,
a.n*p.rate expected_at_psa_rate,a.c-a.n*p.rate modeled_incremental FROM a,p;
```

Interpretation

This is a modeled counterfactual, not individually observed attribution. Its causal interpretation requires valid randomization, consistent measurement and no interference.

ad users	observed ad conversions	psa rate	expected at psa rate	modeled incremental
564577	14423	0.0178541	10080	4342.98

21 / Weekday contrasts with multiplicity

Which weekday contrasts remain after a common correction?

```
SELECT segment weekday,n_psa,n_ad,delta*100 lift_pp,
ci_low*100 ci_low_pp,ci_high*100 ci_high_pp,p_value,holm_p
FROM weekday_effects ORDER BY "order";
```

Interpretation

Intervals are pointwise 95% Newcombe intervals. Segment p-values use two-sided Fisher exact tests, including sparse hourly cells. Holm adjustment controls one exploratory family of 31 weekday/hour contrasts. Conditioning on peak day still limits causal interpretation.

weekday	n psa	n ad	lift pp
Monday	3502	83571	1.06827
Tuesday	2907	74572	1.59925
Wednesday	3490	77418	0.959655
Thursday	3905	79077	0.140666
Friday	3803	88805	0.616203
Saturday	2858	78802	0.731076
Sunday	3059	82332	0.402487

weekday	ci low pp	ci high pp	p value	holm p
Monday	0.508751	1.52724	0.000329804	0.00956431
Tuesday	1.08262	1.99394	6.00701e-08	1.86217e-06
Wednesday	0.477678	1.34001	0.000193963	0.0058189
Thursday	-0.360253	0.550728	0.611393	1
Friday	0.152245	0.986266	0.00971553	0.252604
Saturday	0.22093	1.11526	0.00639199	0.172584
Sunday	-0.1736	0.861815	0.170286	1

22 / Hour contrasts with multiplicity

How much does multiple testing change hour-level evidence?

```
SELECT segment hour,n_psa,c_psa,n_ad,delta*100 lift_pp,
ci_low*100 ci_low_pp,ci_high*100 ci_high_pp,p_value,holm_p
FROM hour_effects ORDER BY segment;
```

Interpretation

Two-sided Fisher exact p-values address within-segment ad-versus-PSA differences, not whether one hour has a larger effect than another. They avoid the small-cell z approximation. Formal interaction diagnostics are separate.

hour	n psa	c psa	n ad
0	227	0	5309
1	187	0	4615
2	181	0	5152
3	89	1	2590
4	28	0	694
5	23	0	742
6	83	0	1985
7	237	2	6168
8	659	7	16968
9	1202	13	29802
10	1485	20	37454
11	2061	30	44149
12	2060	35	45238
13	2170	36	45485
14	1869	30	43779
15	1828	46	42855
16	1604	45	35963
17	1383	28	33605
18	1271	32	31052
19	1183	29	29169
20	1077	19	27846
21	1081	24	28895

hour	n psa	c psa	n ad
22	917	15	25515
23	619	8	19547

hour	lift pp	ci low pp	ci high pp	p value	holm p
0	1.92127	0.223575	2.32678	0.0220193	0.484426
1	1.34345	-0.690818	1.71838	0.17619	1
2	0.756988	-1.33112	1.03309	0.642712	1
3	-0.0811245	-5.06139	0.956432	0.6136	1
4	1.58501	-10.4995	2.81574	1	1
5	2.15633	-12.1791	3.47392	1	1
6	2.31738	-2.14343	3.0771	0.258802	1
7	1.00437	-1.19744	1.71827	0.448948	1
8	0.923876	-0.207793	1.51362	0.112064	1
9	0.871358	0.0964219	1.3486	0.0310902	0.621804
10	0.837211	0.0988477	1.33461	0.0281467	0.591081
11	0.791332	0.162133	1.24822	0.0141277	0.324936
12	0.714871	0.0459363	1.21152	0.0382159	0.726102
13	0.847335	0.202898	1.32882	0.0106942	0.267354
14	1.2524	0.558526	1.75701	0.00074242	0.0207878
15	0.468071	-0.370434	1.11413	0.290299	1
16	0.2838	-0.660094	1.00971	0.60455	1
17	0.829158	-0.0734685	1.4759	0.0682246	1
18	0.229302	-0.800723	0.981811	0.725356	1
19	0.229534	-0.832726	0.993211	0.712887	1
20	1.26321	0.269155	1.92839	0.0135175	0.32442
21	0.697293	-0.381324	1.44827	0.195798	1
22	1.00973	-0.0528957	1.68342	0.0577592	1
23	1.00462	-0.248499	1.67755	0.128564	1

23 / Statistical methods and exact outputs

Primary comparison

```
pA = conversions_PSA / n_PSA
pB = conversions_ad / n_ad
d = pB - pA
pool = (conversions_PSA + conversions_ad) / (n_PSA + n_ad)
z = d / sqrt(pool*(1-pool)*(1/n_PSA + 1/n_ad))
p_two = 2 * norm.sf(abs(z))
```

Newcombe interval

Compute 95% Wilson bounds (LA, UA) and (LB, UB) for PSA and ad. Lower difference = d - sqrt((pB-LB)^2 + (UA-pA)^2). Upper difference = d + sqrt((UB-pB)^2 + (pA-LA)^2). No continuity correction is applied.

Primary output	Value
n_psa	23524
n_ad	564577
c_psa	420
c_ad	14423
p_psa	0.0178541
p_ad	0.0255466
delta	0.00769245
ci_low	0.00587342
ci_high	0.00936003
relative_lift	0.430851
risk_ratio	1.43085
rr_low	1.2997
rr_high	1.57524
z	7.37008
p_value	1.70528e-13
wald_low	0.00595093
wald_high	0.00943397
psa_low	0.0162389
psa_high	0.0196267
ad_low	0.0251382
ad_high	0.0259614
fisher_p	1.04707e-14

Primary output	Value
chi2_p	1.70528e-13
ad_share	0.96
srm_96_4_p	0.999788
srm_50_50_p	0
bootstrap_low	0.00592653
bootstrap_high	0.00940879
incremental_ad	4342.98
incremental_low	3316
incremental_high	5284.46
attributable_fraction	0.301115
number_needed_expose	129.998
variance_psa_share	0.944152
impressions_ad	14014701
cpm_break_even_per_unit_margin	0.309888
p99_exposure	202
above_p99_users	5810
above_p99_impression_share	0.129945
smd_exposure	0.00143686

24 / Multiplicity, interaction and resampling

Holm procedure

Use two-sided Fisher exact p-values for all segment contrasts. Sort all 31 weekday/hour p-values ascending. Multiply the ith value by (31-i), with i starting at zero. Take the cumulative maximum and cap at 1, then restore the original order. The pointwise intervals are not multiplicity-adjusted.

Interaction diagnostics

For each category variable, fit an aggregated binomial logistic model with intercept, treatment indicator and category dummy variables. Compare its likelihood with the saturated category-by-treatment model. LR = 2 x (logLik_full - logLik_restricted); degrees of freedom = categories - 1. Two interaction p-values receive a separate Holm adjustment.

Field	LR	df	p	Holm p
peak_day	16.2195	6	0.0126229	0.0252457
peak_hour	29.0996	23	0.177031	0.177031

Resampling

Generate 100,000 independent binomial counts per arm using the observed conversion probabilities and fixed arm sizes. Convert to rates, subtract PSA from ad and take the 2.5th/97.5th percentiles. Seed: 20261101. This is an empirical-Bernoulli sampling check, not a randomization test under the null.

Extreme-exposure sensitivity

Output	Value
n_psa	23274
n_ad	559017
delta	0.0079189
ci_low	0.00616296
ci_high	0.00952146
p_value	9.54078e-15

25 / Planning and economic scenarios

Prospective normal-approximation sample sizes

Let r = n_ad/n_PSA, p0 = observed PSA rate, p1 = p0 x (1 + target relative lift), and pooled = (p0 + r*p1)/(1+r). Required PSA n = ceil([(z_.975*sqrt(pooled*(1-pooled)*(1+1/r)) + z_.8*sqrt(p0*(1-p0)+p1*(1-p1)/r))/(p1-p0)]^2). Round the ad count upward at the chosen ratio. No trial duration is estimated without eligible traffic rates.

Relative lift	Absolute pp	Split	PSA	Ad	Total
10%	0.1785	50/50	90,587	90,587	181,174
10%	0.1785	96/4	47,971	1,151,304	1,199,275
20%	0.3571	50/50	23,703	23,703	47,406
20%	0.3571	96/4	12,731	305,544	318,275
30%	0.5356	50/50	11,003	11,003	22,006
30%	0.5356	96/4	5,982	143,568	149,550

Economic scenario formulas

Modeled incremental conversions = n_ad x (p_ad-p_PSA). Contribution = incremental conversions x assumed margin. Ad cost = recorded ad-arm impressions / 1,000 x assumed CPM. Net = contribution - cost. Confidence limits scale the primary rate interval; they do not include uncertainty in cost, margin or causal assumptions.

Margin	CPM	Cost	Net point	Net low	Net high
10	2	28,029	15,400	5,131	24,815
10	5	70,074	-26,644	-36,914	-17,229
10	10	140,147	-96,717	-106,987	-87,302
25	2	28,029	80,545	54,871	104,082
25	5	70,074	38,501	12,826	62,038
25	10	140,147	-31,572	-57,247	-8,036
50	2	28,029	189,120	137,771	236,193
50	5	70,074	147,076	95,726	194,149
50	10	140,147	77,002	25,653	124,076

Validation

SQL group totals equal Python group totals. Clean users equal distinct IDs. Dashboard cube reconciles to 588,101 users, 14,843 conversions and the complete exposure sum. Welch-style or t-tests on exposure are not substituted for the binary primary outcome. All primary source users remain included.

26 / Final dashboard simulator audit

Four dynamic inputs; two distinct baseline conventions

The final dashboard uses selectable baseline 1.8%, relative lift 20%, power 80% and ad allocation 50% as defaults. The report planning queries retain the unrounded observed PSA baseline. Significance is fixed at 5%, two-sided. The final export is static; the PBIX is interactive.

Changed input	Expected total users
None / defaults	47,014
Power 90%	62,938
Target relative lift 40%	12,798
Ad allocation 96%	315,650
Baseline conversion 2%	42,218

Formula audit

For p0, relative lift L, power W and ad share a: p1=p0*(1+L), r=a/(1-a), pooled=(p0+r*p1)/(1+r). Compute PSA n by the prospective formula in investigation 25 with z(W) replacing z(.8). Ad n=ceil(r*PSA n), total=PSA n+ad n. Ceiling is applied separately to the two arm counts.

Exact native DAX

Required PSA users

```
VAR p0=SELECTEDVALUE(Baseline[Value],0.018)
VAR lift=SELECTEDVALUE('Target lift'[Value],0.20)
VAR pw=SELECTEDVALUE(Power[Value],0.80)
VAR a=SELECTEDVALUE('Ad allocation'[Value],0.50)
VAR p1=p0*(1+lift)
VAR r=DIVIDE(a,1-a)
VAR pb=DIVIDE(p0+r*p1,1+r)
VAR za=NORM.S.INV(0.975)
VAR zb=NORM.S.INV(pw)
VAR n0=ROUNDUP(DIVIDE(POWER(za*SQRT((1+1/r)*pb*(1-pb))+zb*SQRT(p0*(1-p0)+p1*(1-
p1)/r),2),POWER(p1-p0,2)),0)
RETURN IF(p1<1,n0)
```

Required ad users

```
ROUNDUP([Required PSA users]*DIVIDE(SELECTEDVALUE('Ad
allocation'[Value],0.5),1-SELECTEDVALUE('Ad allocation'[Value],0.5)),0)
```

Required experiment users

```
[Required PSA users]+[Required ad users]
```

27 / Simulator sensitivity and interpretation

Curve total users

```
VAR p0=SELECTEDVALUE(Baseline[Value],0.018)
VAR lift=SELECTEDVALUE('Lift curve'[Value],0.20)
VAR pw=SELECTEDVALUE(Power[Value],0.80)
VAR a=SELECTEDVALUE('Ad allocation'[Value],0.50)
VAR p1=p0*(1+lift)
VAR r=DIVIDE(a,1-a)
VAR pb=DIVIDE(p0+r*p1,1+r)
VAR za=NORM.S.INV(0.975)
VAR zb=NORM.S.INV(pw)
VAR n0=ROUNDUP(DIVIDE(POWER(za*SQRT((1+1/r)*pb*(1-pb))+zb*SQRT(p0*(1-p0)+p1*(1-
p1)/r),2),POWER(p1-p0,2)),0)
RETURN IF(p1<1,n0+ROUNDUP(n0*r,0))
```

Final exposure-page interpretation

The page labels frequency patterns as observational, removes subgroup-lift and average-frequency cards, and calls for randomized frequency-cap trials. Statistical primary results are unchanged. The final dashboard, PBIX, source model, report and SQL workbook are versioned together.